

MUSSIE Y. BEYENE

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EDUCATION

University of California, Los Angeles
B.S. in Pre Math of Computation

Los Angeles, California
Expected Graduation, June 2028

EXPERIENCE

MathWorks

Remote

Computer Vision Engineer (Academic Project) – MathWorks Challenge

June 2026 – July 2026

- Selected as 1 of 200 students for the UCLA Engineering Pathway Program (EPP) to architect an automated manufacturing inspection pipeline for the MathWorks Workplace Challenge.
- Engineered a hybrid MATLAB inspection pipeline combining classical computer vision and ResNet-50 deep learning to automate PASS/FAIL defect detection, achieving 98% classification accuracy.
- Built algorithms to isolate defect regions and extract quantitative physical properties (e.g., maximum defect area, component count), generating visual evidence overlays that reduced QA team review time by 95%.
- Fine-tuned a ResNet-50 CNN via transfer learning for binary PASS/FAIL classification of pharmaceutical pill defects, **achieving 97% validation accuracy**.

AlgoVerse

Remote

AI and Machine Learning Intern

June 2024 – Sept 2025

- Designed and built the SIOR Benchmark, generating syntactically perturbed math problem variants to test LLM reasoning robustness across 10 problem types, uncovering significant performance degradation in leading models.
- Evaluated GPT-4, Claude 3 Opus, LLaMA 3, and Mistral using Chain-of-Thought prompting and custom metrics (Linguistic Fragility Index, validity scores, error classification), finding that all models showed >90% performance variance across reformulations.

Santa Monica College

Santa Monica, California

Peer Mentor / Tutor

May 2024 – Present

- Supported 120+ students weekly across C, Java, Python, C++, Pre-Calculus, Calculus 1–3, and Mechanics, helping the majority achieve passing grades in their respective courses.
- Created structured, adaptive lesson plans for 10+ students per semester, incorporating visual aids and problem-solving frameworks that improved comprehension of complex CS and math topics.
- Guided students through challenging CS and math problem sets using Socratic questioning and step-by-step breakdowns, helping 5+ students pass previously failed courses.

PROJECTS & OUTSIDE EXPERIENCE

SMC Transfer Tracker - Transfer Planning Web Application
Santa Monica, California

Oct 2025 – Present

- Built a responsive SPA with React, TypeScript, and Tailwind CSS that enables SMC students to generate personalized transfer roadmaps with drag-and-drop planning, real-time prerequisite validation, and progress tracking across 10+ majors and UC/CSU universities — used by 20+ students.
- Designed a JSON-based data layer integrating SMC course catalog, ASSIST.org articulation agreements, and IGETC requirements, with prerequisite chain validation and automated conflict alerts to prevent invalid course selections.

ACTIVITIES & LEADERSHIP

ColorStack at SMC

Santa Monica, California

President

Aug 2019 – Current

- Founded and managed ColorStack chapter supporting underrepresented students in computer science through technical workshops and industry partnerships, coordinating 5+ events per semester with guest speakers from different companies
- Facilitated peer study groups for core CS courses, managed chapter communications via Slack/Discord with 50+ active members, and collaborated with SMC administration to secure funding and campus resources for chapter activities

SKILLS & INTERESTS: Java, Python, JavaScript, HTML/CSS, C++, RAG, MATLAB, Computer Vision